

W1: NLP (Natural Language Processing, 自然语言处理) Overview

★ Quiz Key Points

Transformer 2017: C – self-attention, turning point

Sentiment complexity: True – sarcasm/irony/metaphor

NLG example: D – Text Classification is NLU not NLG

Document vs Knowledge: True – raw text vs structured info

Variation/Zipf: True – skewed distribution

AI/ML/DL/NLP: D – AI>ML>DL; NLP=AI app domain

Summarization: False – MUST preserve coherence

NLP goal: True – understand/interpret/generate

AI fixed rules: False – AI includes learning

Turing Test: True – evaluator can't distinguish

★★★★ Core Definitions (核心定义)

Term	Definition
NLP (Natural Language Processing, 自然语言处理)	Linguistics × CS × AI
NLU (Natural Language Understanding, 自然语言理解)	Classification, NER, Sentiment Analysis
NLG (Natural Language Generation, 自然语言生成)	Translation, Summarization, ChatGPT
NLP = NLU + NLG	Understanding + Generation

▲ Text Classification (文本分类) = NLU (NOT NLG)

▲ Sentiment Analysis (情感分析) requires deep contextual understanding (sarcasm/irony), not keyword matching

▲ Summarization (文本摘要) MUST preserve meaning and coherence

▲ "AI relies only on fixed rules" → False

▲ "ML is unrelated to AI" → False

★★★★ AI Hierarchy (AI层级关系)

AI (Artificial Intelligence, 人工智能) > ML (Machine Learning, 机器学习) > DL (Deep Learning, 深度学习)

NLP is an application domain of AI, spans all levels

★★★★ 7 NLP Applications (7大NLP应用)

Application	Type
Speech Recognition (语音识别)	NLU
Dialogue/Chatbot (对话/聊天机器人)	NLU+NLG
Text Classification (文本分类)	NLU
Sentiment Analysis (情感分析)	NLU
Summarization (文本摘要)	NLG
QA (Question Answering, 问答系统)	NLU+NLG
Generative AI (生成式AI)	NLG

★★★★ Ambiguity Types (歧义类型)

Type	Example
Lexical (词汇歧义)	'bank' = financial institution OR river edge; 'bass' = fish OR low sound
Syntactic (句法歧义)	'She saw the man with the telescope' – who has it?
Referential (指代歧义)	'My girlfriend and I met my lawyer... she had to leave' – who is 'she'?

★★★★ 4 NLP Challenges (4大NLP挑战)

Challenge	Definition
Ambiguity (歧义性)	Lexical (bank), Syntactic (attachment), Referential (pronoun)
Sparsity (稀疏性)	Zipf's Law: freq ∝ 1/rank, >1/3 words appear once
Variation (变异性)	Lexical/Syntactic/Regional/Social/Stylistic/Generational
Common Knowledge (常识知识)	Machines lack world knowledge

★★★★ NLP Motivations (NLP动机)

Motivation

1 Enable computers to understand/process human language

2 Help businesses analyze customer feedback & improve service

3 Make it easier for individuals with disabilities to interact with computers

4 Extract insights and make sense from data

★★★★ Knowledge Representation (知识表示) – Comparison

Document (文档) | Raw/semi-structured text

Knowledge (知识) | Structured, interpreted info (entities, relations, facts)

Structured (结构化) | Tables/databases

Unstructured (非结构化) | Free text, >80% enterprise data

Dimension	Structured Knowledge (结构化)	Unstructured (非结构化)
Nature	Precise, Actionable	Ambiguous
Scope	Specific to task	Lots of it (>80% enterprise data)
Humans	Can read, but slowly	Can't remember all
Computers	Use quickly, answer questions	Need NLP to process

★★★★ 3 NLP Approaches (3种NLP方法)

Approach	Method
Heuristics (启发式)	Rules / Regex
ML (Machine Learning, 机器学习)	Supervised / Unsupervised
DL (Deep Learning, 深度学习)	RNN, LSTM, Transformer

★★★★ Turing Test (图灵测试, 1950)

Human evaluator interacts via text with one human + one machine

If evaluator cannot distinguish → machine passes

W2: Text Preprocessing (文本预处理)

★ Quiz Key Points

Tokenization: A – raw text → processable units

No stem/lemma: B – Poetry Analysis

Spacy stemming: True – no built-in stemming

Regex hyphen: False – compound words, not ending hyphen
Text cleaning: True – removes noise, improves quality
Vowel check: B – [s0] and [s-1] in 'aeiou'
Lemma vs stem: A – accuracy over speed
Stemming action: B – helped/helps-help
Regex d+: False – multiple matches, not just 'and'

★★★★ Stemming vs Lemmatization (词干提取 vs 词元化)

Dimension	Stemming (词干提取)	Lemmatization (词元化)
Method	Rule-based suffix stripping	Dictionary lookup + POS
Speed	⚡ Faster	🐢 Slower
Accuracy	Lower (may produce 'studi')	🟢 Higher (valid dictionary words)
When to use	Speed priority	Accuracy priority
Example	helped-helps-help	was/am/is-be, better-good

▲ helped/helps-help = Stemming (suffix removal -ed, -s)
▲ was/am/is-be = Lemmatization (dictionary knowledge for irregular forms)

▲ Poetry Analysis: do NOT stem/lemmatize (rhyme/rhythm crucial)
▲ Spacy has NO Stemming; NLTK has both

★★★★ Spacy vs NLTK (工具对比)

Library	Stemming	Lemmatization
Spacy	❌ No	✅ Yes
NLTK	✅ PorterStemmer/SnowballStemmer	✅ WordNetLemmatizer

★★★★ When NOT to Stem/Lemmatize (何时不用词干/词元化)

Scenario	Reason
Poetry Analysis (诗歌分析)	Rhyme/rhythm crucial
Social Media (社交媒体)	Slang, abbreviations, intentional misspellings
Computational Cost (计算成本)	Speed matters more than normalization

★★★★ Python re Module Functions (re模块函数)

Function	Description
match(r,s)	Start only
search(r,s)	First anywhere
findall(p,s)	All-list
sub(p,repl,s)	Replace
compile(p)	Compile pattern
split(p,s)	Split by pattern
groups()	All group substrings tuple
span()	Match (start,end) position

▲ \b[w+](\w+)*b → compound words (high-tech), NOT words ending with hyphen
▲ [a-zA-Z]\w+ → multiple matches, not just 'and'
▲ s[0]=first, s[-1]=last; vowel check: s[0].lower() in 'aeiou' and s[-1].lower() in 'aeiou'

★★★★ Preprocessing Pipeline (预处理流水线)

- Documents → Tokenization (分词) → Noise Removal (噪声移除) → Normalization (规范化)
- Tokenization = FIRST step in NLP pipeline
- Corpus (语料库) = collection of text documents

★★★★ NLTK Stemmers (NLTK词干提取器)

Stemmer	Characteristic
Porter Stemmer	Most common, least aggressive
Snowball Stemmer	Improved Porter, multi-language
Lancaster Stemmer	Most aggressive, may over-stem

★★★★ POS (Part-of-Speech, 词性) Tagging (词性标注)

- Why: syntactic/semantic analysis, improve other NLP tasks
- nlk.help.upenn_tagset()

Tag	Meaning
NN	Noun
VB	Verb
JJ	Adjective
RB	Adverb
DT	Determiner
PRP	Pronoun

★★★★ NER (Named Entity Recognition, 命名实体识别)

- Why: information extraction, searching/indexing, sentiment analysis
- from nltk.chunk import ne_chunk

Entity Type	Example
PERSON	Names of people
LOCATION/GPE	Countries, cities
ORGANIZATION	Companies, institutions
DATE	Dates, time expressions
MONEY	Monetary values

★★★★ Regex Metacharacters (正则元字符)

Symbol	Meaning
.	Any single char
[]	Char set
[^]	Negate
^	Line start
\$	Line end
*	0+ times
+	1+ times
?	0 or 1
{m,n}	m to n times
\	Escape
	OR
()	Capture group

★★★★ Regex Character Classes (正则字符类)

Class	Meaning	Negation
\d	[0-9]	\D
\w	[A-Za-z0-9_]	\W
\s	Whitespace	\S

W3: Text Vectorization (文本向量化) & Similarity (相似度)

★ Quiz Key Points

- Edit distance: C=5 – intention→execution
- BOW order: False – ignores word order
- IDF formula: True – N/df(t)
- Cosine=1: True – semantically similar
- TF-IDF limitation: C – no context/semantics
- TF sorting: False – d1 and d9 swapped
- Cosine calc: ≈0.8421
- CountVectorizer: False – 8+ features not 7

★★★★ BOW (Bag-of-Words, 词袋模型)

- BOW word order loss: 'John is quicker than Mary' = 'Mary is quicker than John' → same vector!

Advantages	Disadvantages
Simplicity & Interpretability (简单可解释)	Ignores context & word order (忽略上下文和词序)
Works for text classification & IR (适合分类和检索)	High-dimensional feature space (高维特征空间)
Computational Efficiency (计算效率高)	Sparsity (稀疏性)
Language Agnostic (语言无关)	Lack of semantic info (缺乏语义)
	OOV (Out-of-Vocabulary, 未登录词)

★★★★ N-gram (N元组)

- Unigram='I', 'am', 'learning', 'NLP' | Bigram='am learning', 'learning NLP' | Trigram='am learning NLP'

Advantages	Disadvantages
Captures partial context/word order (捕获部分上下文)	Sparsity (稀疏性) – vocab grows as V ⁿ
Simple & efficient (简单高效)	Computationally expensive (计算开销大)
	Ignores overall structure/meaning (忽略整体语义)
	Choice of N is hard (N的选择困难)
	OOV (Out-of-Vocabulary, 未登录词)

★★★★ TF-IDF (Term Frequency-Inverse Document Frequency, 词频-逆文档频率) – Formulas

- Word in every doc: IDF=log(1)=0 → zero weight
- Word in 1 doc: IDF=log(N) → highest weight
- Why log: compress extreme values to manageable range

$$TF(t,d) = \text{count}(t \text{ in } d) / \text{total words in } d$$

$$IDF(t) = \log(N / df(t))$$

$$TF-IDF(t,d) = TF(t,d) \times IDF(t)$$

★★★★ TF-IDF – 5 Limitations (5大局限)

#	Limitation
1	Miss word relationships (缺失词间关系)
2	Sparsity (稀疏性)
3	Lack of semantic understanding (缺乏语义理解)
4	Not suited for small corpora (不适合小语料库)
5	OOV (Out-of-Vocabulary, 未登录词)

★★★★ CountVectorizer vs TfidfVectorizer (计数向量器 vs TF-IDF向量器)

- CountVec: doc1('hot sun') & doc3('hot encoding') most similar (0.408) – misleading!
- TF-IDF: doc2('chocolate milk') & doc4('latte milk') most similar (0.232) – smarter!
- ngram_range=(1,2) | Unigrams & bigrams
- Vector dim | Total n-grams in vocabulary

Property	CountVectorizer	TfidfVectorizer
Output	Raw word counts	TF-IDF weighted scores
Common words	High count (misleading)	Low weight (downweighted)
Rare words	Low count	High weight (upweighted)
Example	'hot' in 4/5 docs → high count	'milk' in 2/5 docs → higher TF-IDF

▲ Corpus with (1,2)-gram produces 8+ features, NOT 7

★★★★ Traditional Methods Comparison (传统方法总对比)

Dimension	OHE (独热编码)	BOW (词袋模型)	N-Gram (N元组)	TF-IDF (词频-逆文档频率)
Frequency	❌ No	✅ Yes	✅ Yes	✅ Weighted
Word order	❌ No	❌ No	✅ Partial	❌ No
Semantics	❌ No	❌ No	❌ No	❌ No
Sparsity	Very high	High	Very high (V ⁿ)	High
Rare words	❌	❌	❌	✅ Upweights

▲ BOW IGNORES word order

▲ TF-IDF does NOT consider context/semantic relationships

▲ TF-IDF = high-dim sparse; DL prefers low-dim dense

★★★★ Similarity Measures (相似度度量) – Formulas

Measure	Formula/Range	Length-independent?
Cosine (余弦相似度)	$\cos(\theta) = \frac{A \cdot B}{ A \times B }$, [-1,1]	✅ Yes
Euclidean (欧几里得距离)	$d = \sqrt{(a_1 - b_1)^2}, [0, \infty)$	❌ No
Levenshtein (编辑距离)	Min insert/delete/substitute, [0, ∞)	N/A

★★★★ Cosine Similarity (余弦相似度) – Example

$$w_1 = (0.2, 0.2, 0.3, 0.7), w_2 = (0.3, 0.4, 0.8, 0.5)$$

$$\text{Dot} = 0.06 + 0.08 + 0.24 + 0.35 = 0.73$$

$$||w_1|| = \sqrt{0.66} = 0.8124$$

$$||w_2|| = \sqrt{1.14} = 1.0677$$

$$\cos(\theta) = 0.73 / (0.8124 \times 1.0677) = 0.8421$$

★★★★ Edit Distance (编辑距离) – Examples

Transformation	Operations
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kitten → sitting 3 ops (2 substitutions + 1 insertion)

intention → execution 5 steps (Dynamic Programming)

★★★★ VSM (Vector Space Model, 向量空间模型)

Concept	Definition
Vector Space	Text → points in vector space; similar words nearby
Vectorizing	Encoding text as integers → feature vectors

$$L_2 \text{ Norm } ||x|| = \sqrt{\sum x_i^2}$$

$$\text{Dot Product } x_1 \cdot x_2 = \sum x_{1i} x_{2i}$$

★★★★ OHE (One-Hot Encoding, 独热编码)

Property	Detail
Method	Each word → binary vector, only one 1, rest 0
✅ Advantage	Simple to implement
❌ Sparse	Vector size = vocab size
❌ No semantics	Cannot capture word meaning
❌ No frequency	All words weighted equally

★★★★ 5 Similarity Measures (5种相似度度量)

#	Measure
1	Jaccard Similarity (Jaccard相似度)
2	Cosine Similarity (余弦相似度)
3	Euclidean Distance (欧几里得距离)
4	Hamming Distance (汉明距离)
5	Levenshtein Distance (编辑距离)

W4: Word Embeddings (词嵌入)

★ Quiz Key Points

TF-IDF for DL: True – high-dim sparse, bad for DL
Word analogy: e_boy-e_girl=e_brother-e_sister
Self-supervised: True – no manual labels
GloVe global+local: True – co-occurrence + window
Embedding dim: False – 50-300, not vocab size
Gensim default: 100
W2V architectures: True – CBOW + Skip-gram
Skip-gram direction: False – center+context (not reverse)
Modern NLP embeddings: False – all modern NLP uses embeddings

★★★ WordNet (词汇数据库)

Limitations: limited coverage, static, not computational, domain-specific, manual curation

Relation	Definition	Example
Synset (同义汇集)	Synonyms sharing meaning	{car, auto, automobile}
Hypernym (上位词)	IS-A general	animal>dog
Hyponym (下位词)	IS-A specific	dog>animal
Meronym (部分词)	PART-OF	wheel>car
Holonym (整体词)	HAS-PART	car>wheel
Antonym (反义词)	Opposite	hot>cold
Troponym (方式动词)	Manner of verb	run>move
Entailment (蕴含)	Implies	snore>sleep

▲ Hypernym/Hyponym = IS-A | Meronym/Holonym = PART-OF – don't confuse!

★★★ Word Embedding Fundamentals (词嵌入基础)

Distributional Hypothesis (分布假说): similar contexts → similar meanings
Word Embedding: dense low-dim vector (d=50-300)
Self-Supervised Learning (自监督学习): no manual labels
Word Analogy: king-man+woman=queen; e_boy-e_girl=e_brother-e_sister
▲ 1000-word vocab + 1000-dim embedding? → False (typical 50-300)
▲ "Modern NLP doesn't use embeddings" → False (W2V+GloVe+BERT+GPT all use)

★★★ TF-IDF vs Word Embeddings (TF-IDF vs 词嵌入)

1 | High-dimensional representation (高维表示)

2 | Sparse (稀疏)

3 | OOV words (未登录词)

4 | Lack of semantic meaning (缺乏语义含义)

5 | → Poor performance on majority of NLP tasks

Property	TF-IDF (词频-逆文档频率)	Word Embeddings (词嵌入)
Dimensionality	High (=vocab size)	Low (50-300)
Sparsity	Sparse	Dense
Semantics	None	✓ Captures meaning
Context	None	✓ Learned
DL suitability	✗ Poor	✓ Good

★★★ CBOW (Continuous Bag-of-Words, 连续词袋) vs Skip-gram (跳字模型)

Gensim default: vector_size=100
sg=1: Skip-gram | sg=0: CBOW | window=5 | negative=5

Property	CBOW (连续词袋)	Skip-gram (跳字模型)
Direction	Context-Center word	Center word-Context
Speed	⚡ Faster	🐢 Slower
Best for	Frequent words, large corpus	Rare words, small corpus
Difficulty	Easier	Harder

▲ "Skip-gram predicts center from context" → False (that's CBOW!)

★★★ SGNS (Skip-Gram Negative Sampling, 跳字负采样)

1. Target+context = positive | 2. Random words = negative
3. Logistic regression classifier | 4. Weights = embeddings
Reduces softmax O(V) → binary O(k)

★★★ Word2Vec vs GloVe (Global Vectors, 全局词向量) vs FastText

Property	Word2Vec	GloVe (全局词向量)	FastText (快速文本)
Method	Prediction	Count+factorization	Prediction+subword
Data	Local context	Global co-occurrence	Local+char n-grams
OOV	✗	✗	✓ Subword
Year	2013	2014	2016

★★★ Word Similarity & Semantic Field (词相似度与语义场)

Concept	Detail
Word Similarity	Cat ≠ synonym of dog, but similar words
Semantic Field (语义场)	Set of words covering a semantic domain
Restaurant field	waiter, menu, plate, food, chef
House field	door, roof, kitchen, family, bed

★★★ FastText Subword (子词) Detail

Property	Detail
Char n-grams	Length 3-6 for each word
Center word embedding	sum(char n-gram vectors + whole word vector)
Context words	Use word vector directly (no char n-grams)
Advantages	Fine-grained info, solve OOV, open-source, multi-language

★★★ Embedding Benefits & Limitations (词嵌入优缺点)

Benefits	Limitations
Dimensionality reduction	Context insensitivity (static)
Semantic meaning	Bias in training data
OOV handling (FastText)	Limited semantic adaptation
Transfer learning	Resource intensive

★★★ Embedding Evaluation (词嵌入评估)

Type	Description
Intrinsic (内在)	Word similarity, analogy tasks
Extrinsic (外在)	Downstream: classification, NER

★★★ Universal Text Representations (通用文本表示) – Preview

Property	Detail
Key idea	Contextual word representations (上下文词表示)
Models	ELMo, BERT, ULMFiT
Method	Multiple passes: left-to-right AND right-to-left
Advance	Advanced neural language models (高级神经网络语言模型)
Significance	Next generation beyond static embeddings

W5: Language Models (语言模型) & RNN/LSTM

★ Quiz Key Points

- RNN gradient: True – $\partial L / \partial \theta$ via BPTT
- Stateful computation: True – h_t across time steps
- Training data: True – diverse public datasets
- Bigram prob: $B=0.4$ – $\text{Count}(AB) / \text{Count}(A)$
- RNN vs FFN: A – internal state for sequential deps
- LSTM gates: C – gating regulates info flow
- N-gram semantics: False – statistical only, no deep reasoning
- Low LR: False – slow convergence, not faster

★★★ Gradient Problems (梯度问题)

Problem	Cause	Effect
Vanishing (梯度消失)	Gradient=0	Can't learn long-range deps
Exploding (梯度爆炸)	Gradient→∞	Unstable training

★★★ LSTM (Long Short-Term Memory, 长短期记忆网络)

Markov Assumption: only previous n-1 words
Limitations: sparsity, fixed window, no semantics
Limitation

1	Data Sparsity (数据稀疏性)
2	Computational Complexity (计算复杂度)

★★★ PPL (Perplexity, 困惑度) & LR (Learning Rate, 学习率)

PPL: low=good prediction, high=confused Setting Effect LR too low SLOW convergence, local optima LR too high Oscillation/divergence	
▲ "Low LR → faster training" → False (OPPOSITE!)	
★★★ UAT (Universal Approximation Theorem, 通用逼近定理)	
Property	Detail
Author	Cybenko 1989
Theorem	NN with single sufficiently wide hidden layer can approximate any continuous function

★★★ Perceptron (感知器)

Component	Detail
Input	$x_1...x_n$
Weights	$w_1...w_n$
Sum	$v = \sum (w_i x_i) + b$
Output	$y = \text{activation}(v)$
Role	Basic building block of neural networks

★★★ Fixed-window NN LM (固定窗口神经网络语言模型)

Property	Detail
Architecture	one-hot → embedding → concat → hidden → softmax
✓ Advantage	Embeddings capture semantics
✗ Limitation	Fixed window size

▲ FFNN Limitation: 'good, not bad' vs 'bad, not good' look similar in small window	
★★★ Sequence Modeling (序列建模) – Motivations	
# Motivation	
1 Handle variable-length sequence data (处理变长序列)	
2 Track long-term dependency (追踪长期依赖)	
3 Maintain order information (维护顺序信息)	
4 Share information across sequence (序列中共享信息)	

★★★ LM Evolution (语言模型演化)

Stage	Model	Key Feature
1	N-gram	Statistical counting
2	FFNN	Embeddings, fixed window
3	RNN	Variable-length, param sharing
4	LSTM	Gating solves vanishing gradient

W6: Seq2Seq (序列到序列) & Attention (注意力)

★★★ Bi-LSTM (Bidirectional LSTM, 双向长短期记忆网络)

Property	Detail
Structure	Forward (left→right) + Backward (right→left), concat: $h_t = [h_t, \bar{h}_t]$
Output dim	$2 \times n_lstm$, separate weights
✗ Cannot	Generate text
✓ Good for	Classification, NER, sentiment
Motivation	'terribly exciting' – forward sees 'terribly'→negative? Need right context 'exciting'→positive!

★★★ 5 Sequence Architectures (5种序列架构)

Type	I/O	Example
One-to-one	Fixed-Fixed	Image classification
One-to-many	Fixed-Seq	Image captioning
Many-to-one	Seq-Fixed	Sentiment analysis
Many-to-many sync	Seq-Seq same	POS tagging
Many-to-many unsync	Seq-Seq diff	Translation (Seq2Seq)

★★★ Seq2Seq (Sequence-to-Sequence, 序列到序列) / Encoder-Decoder

Component	Detail
Encoder	LSTM reads source → fixed vector
Decoder	Uses vector as init state, autoregressive generation
Formulation	Conditional LM: $P(y_1...y_t x_1...x_n)$
Loss	$J = (1/T) \sum J_t$, end-to-end backprop

★★★ Seq2Seq Training vs Testing (训练 vs 测试)

Phase	Decoder Input	Behavior
Training	Ground truth (Teacher Forcing)	Fast, stable, parallel-friendly
Testing	Own previous prediction (Autoregressive)	Error accumulation, sequential

▲ Exposure Bias (曝光偏差): model never sees its own errors during training → fragile at test time

★★★ Attention (注意力机制) – Definition & Steps

- 1. score_i = dot(decoder_state, encoder_state_i)
- 2. a = softmax(scores) → attention distribution
- 3. context = $\sum a_i \times \text{encoder_state_i}$ → attention output
- 4. [decoder_state ; context] → concat → prediction

Step	Detail
Definition	Attention = weighted average over a set of inputs
1. Similarity	Compute pairwise similarity: each encoder state vs decoder state
2. Distribution	Convert scores to probability distribution (softmax)
3. Weighted avg	Compute weighted average of encoder hidden states

Translation: 'il a m' entarté' → 'he hit me with a pie'
 Decoding 'il': attention focuses on 'il' (highest a)
 Decoding 'he': attention shifts to 'a'
 Decoding 'me': attention on 'm'
 Decoding 'pie': attention on 'entarté'
 Each step: score=softmax-weighted sum=concat-predict

★★★ Seq2Seq vs Seq2Seq+Attention (对比)

- Problem | Single fixed-length vector must capture ALL source info
- Effect | Long sentences → information loss
- Solution | → Attention mechanism

Feature	Seq2Seq	+Attention
Encoding	Single fixed vector	Dynamic weighted sum
Long sentences	✗ Info loss	✓ Attends relevant
Interpretability	✗ Black box	✓ Alignment viz
Gradient	Through bottleneck	✓ Shortcut paths

★★★ Transformer (2017, 变换器)

Property	Detail
Paper	'Attention Is All You Need' (Vaswani et al.)
Self-Attention	Each word attends to ALL words, fully parallel
Positional Encoding (位置编码)	Explicit position info (no recurrence)
Path length	O(1) between any two positions
Architecture	Encoder stack + Decoder stack
Components	Multi-Head Attention, FFN, LayerNorm, Residual Connections

★★★ BERT (Bidirectional Encoder Representations from Transformers, 双向编码器) vs GPT (Generative Pre-trained Transformer, 生成式预训练)

Property	BERT (Bidirectional Encoder Representations from Transformers)	GPT (Generative Pre-trained Transformer)
Architecture	Encoder only	Decoder only
Direction	Bidirectional (both sides)	Autoregressive (left-to-right)
Task	Understanding (NLU)	Generation (NLG)
Pre-training	Masked LM + Next Sentence	Next token prediction
Use cases	Classification, NER, QA	Text generation, chatbot

★★★ Architecture Evolution (架构演化)

From-To	Solved	Trade-off
LSTM-Bi-LSTM	Direction blindness	Can't generate; 2x params
Bi-LSTM+Seq2Seq	Diff I/O lengths	Bottleneck
Seq2Seq+Attention	Bottleneck	Still sequential
RNN+Attn+Transformer	Sequential+parallel	O(n²); needs pos encoding

★★★ Unidirectional LSTM vs Bi-LSTM (单向 vs 双向)

Feature	Unidirectional LSTM	Bi-LSTM
Context	Left only	Both sides
Output dim	n_lstm	2xn_lstm
Generation	✓ Can	✗ Cannot
Understanding	Good	✓ Better

★★★ Multi-layer RNN/LSTM (多层循环神经网络)

Property	Detail
Connection	Hidden states from layer i → inputs to layer i+1
Benefit	Captures hierarchical features
Depth	Deeper = more abstract representations